Light

GCSE Computer Science — Data Representation: Numbers (Set 3)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Why do computers use binary to store and process data?

- ○ **A.** Binary is faster to calculate than denary
- ○ **B.** Electronic circuits can easily represent two states: on and off
- ○ **C.** Binary uses less memory than other number systems
- ○ **D.** Humans find binary easier to understand

Q2 (1 mark)

In binary, what is the place value of the third bit from the right?

- ○ **A.** 2

- B. 3
- C. 8
- D. 4

Q3 (1 mark)

What is the denary number 6 in binary?

- A. 0101
- B. 0110
- C. 0111
- D. 1000

Q4 (1 mark)

What is the binary number 1101 in denary?

- A. 11
- B. 12
- C. 13
- D. 14

Q5 (1 mark)

What denary value does the hexadecimal digit F represent?

- A. 10
- B. 12
- C. 14
- D. 15

Q6 (1 mark)

A programmer is examining a memory address displayed as 7F.
What number system is being used?

- A. Binary
- B. Denary
- C. Hexadecimal
- D. Octal

Q7 (1 mark)

How many different values can be represented using 5 bits?

- A. 5
- B. 10
- C. 25
- D. 32

Q8 (1 mark)

What is the result of adding the binary numbers 0100 and 0011?

- ○ **A.** 0101
- ○ **B.** 0110
- ○ **C.** 1000
- ○ **D.** 0111

Q9 (1 mark)

The binary number 00000011 is shifted one place to the left.
What is the result?

- ○ **A.** 00000001
- ○ **B.** 00000110
- ○ **C.** 00000111
- ○ **D.** 00001100

Q10 (1 mark)

The binary number 00101000 is shifted one place to the right.
What is the result?

- ○ **A.** 00010100
- ○ **B.** 01010000
- ○ **C.** 00100100

- **D.** 00101001

Q11 (1 mark)

What is the term for when a calculation produces a result too large to store in the available bits?

- **A.** Underflow
- **B.** Overflow
- **C.** Truncation
- **D.** Rounding

Q12 (1 mark)

In 8-bit two's complement, what denary value does 10000000 represent?

- **A.** -1
- **B.** -127
- **C.** -128
- **D.** 128

Q13 (2 marks)

Convert the denary number 99 to an 8-bit binary number.

- **A.** 01100001

- B. 01100011
- C. 01100101
- D. 01100111

Q14 (2 marks)

Convert the 8-bit binary number 10100011 to denary.

- A. 159
- B. 161
- C. 163
- D. 165

Q15 (2 marks)

Convert the binary number 11100101 to hexadecimal.

- A. E3
- B. E5
- C. E7
- D. E9

Q16 (2 marks)

Convert the hexadecimal number 9A to binary.

- A. 10011001
- B. 10011010
- C. 10011011
- D. 10011100

Q17 (2 marks)

Calculate the result of adding the binary numbers 01011010 and 00100111.

- A. 10000001
- B. 10000011
- C. 01111111
- D. 01111101

Q18 (2 marks)

The binary number 00001001 is shifted two places to the left.
What denary value does the result represent?

- A. 18
- B. 27
- C. 36
- D. 45

Q19 (2 marks)

The binary number 01110000 is shifted three places to the right.
What denary value does the result represent?

- ○ **A.** 7
- ○ **B.** 14
- ○ **C.** 28
- ○ **D.** 56

Q20 (2 marks)

A student attempts to store the result of adding 150 + 120 in an unsigned 8-bit register.
What value would actually be stored?

- ○ **A.** 14
- ○ **B.** 270
- ○ **C.** 255
- ○ **D.** 0

Q21 (2 marks)

In 8-bit two's complement, what is the binary representation of -19?

- ○ **A.** 11101100
- ○ **B.** 11101101

- **C.** 11101110
- **D.** 11101111

Q22 (2 marks)

Convert the hexadecimal number 3F to denary.

- **A.** 57
- **B.** 59
- **C.** 61
- **D.** 63

Q23 (2 marks)

A graphics system uses 10 bits to store the brightness level of each pixel. How many different brightness levels are possible?

- **A.** 100
- **B.** 512
- **C.** 1000
- **D.** 1024

Q24 (3 marks)

Two unsigned 8-bit binary numbers are added together:
11010101 + 01010011
What is the correct denary result, and what problem occurs?

- ○ **A.** Result is 296, and overflow occurs because this exceeds 255
- ○ **B.** Result is 40, and no problem occurs
- ○ **C.** Result is 296, and no problem occurs
- ○ **D.** Result is 255, and the value is capped at maximum

Q25 (3 marks)

Convert the denary number -100 to an 8-bit two's complement binary number.

- ○ **A.** 10011011
- ○ **B.** 10011100
- ○ **C.** 10011101
- ○ **D.** 10011110

Q26 (3 marks)

Convert the denary number 241 to hexadecimal.

- ○ **A.** EF
- ○ **B.** F0
- ○ **C.** F1
- ○ **D.** F2

Q27 (3 marks)

A signed 8-bit two's complement number 11100000 undergoes an arithmetic right shift of three places.
What is the result in denary?

- ○ **A.** -4
- ○ **B.** -8
- ○ **C.** -16
- ○ **D.** -32

Q28 (3 marks)

A student performs the following binary calculation:
00110010 + 00101000 + 00011110
What is the final result in denary?

- ○ **A.** 116
- ○ **B.** 118
- ○ **C.** 120
- ○ **D.** 122

Q29 (3 marks)

The 8-bit binary pattern 11100110 can be interpreted as either an unsigned integer or a signed two's complement integer.
What are the two possible denary values?

- ○ **A.** Unsigned: 230, Signed: -26

- B. Unsigned: 230, Signed: -230
- C. Unsigned: 26, Signed: -230
- D. Unsigned: 26, Signed: -26

Q30 (3 marks)

An RGB colour value is represented in hexadecimal as #C4A. Assuming this is a 12-bit value (4 bits per hex digit), what is the equivalent value in denary?

- A. 3114
- B. 3130
- C. 3146
- D. 3162

Total: **55** marks • Time: **30 min**

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